PHARMACOGNOSY · PRACTICAL

Pharmacognosy Practical
MCU Examination Study Guide

A complete laboratory revision resource — Second Semester

Microscopy · Crude Drugs · Starches · Gums · Fibres & Dressings · Macroscopy · Diagnostic Characters

An original study guide by the EverythingABUAD team.

everythingabuad.com

Table of Contents

Chapter 1: Introduction to Practical Pharmacognosy	2
Chapter 2: Use and Care of the Microscope	3
Chapter 3: Practical Techniques in Pharmacognosy	5
Chapter 4: Diagnostic Structures of Plant Material	6
Chapter 5: Macroscopy of Crude Vegetable Drugs	8
Chapter 6: Study of Starches (Schedule II)	10
Chapter 7: Unorganised Crude Drugs: Gums (Schedule III)	11
Chapter 8: Examination of Fibres and Surgical Dressings (Schedule I)	12
Chapter 9: Diagnostic Characters of Medicinal Plants	13
Chapter 10: Vernonia amygdalina (Bitter Leaf)	14
Chapter 11: Mineral Matters (Schedule IV)	15
Chapter 12: Quick Revision — Key Facts for the MCU	17

How to Use This Study Guide

This guide pulls together the full second-semester Pharmacognosy practical syllabus into a single, readable revision resource. Every chapter has been rewritten from our own notes into clear explanations and clean comparison tables so you can revise a topic completely from this document alone. Work through each chapter in order, then use the quick-revision chapter at the end to self-test before your MCU. Pay special attention to the results tables and the botanical names, families, and diagnostic features — those are where marks are won and lost.

Before you begin

Read each specimen's **organoleptic characters** (colour, feel, taste), its **microscopic diagnostics** (crystals, stomata, trichomes), and its **chemical test results** together. Examiners like to ask you to tell two similar drugs apart — so learn the *differences*, not just the descriptions.

Chapter 1 • Introduction to Practical Pharmacognosy

Pharmacognosy is the study of medicines that come from natural sources. The drugs and excipients we work with in the laboratory are drawn from three broad origins — plant, animal, and mineral. In their raw, largely unprocessed form these materials are called **crude drugs**.

What are crude drugs?

Crude drugs are natural substances obtained from plants, animals, or minerals that have not been put through extensive processing — at most they are dried or given some simple preparatory treatment. They are the raw starting materials from which medicines and other pharmaceutical products are made.

The three aspects of practical pharmacognosy

Practical work in this course is built around three complementary lines of investigation:

- **Botanical aspects** — naming (nomenclature) and the study of a plant drug's external (macro-morphological) and internal (micro-morphological) structure. This involves sectioning, staining, simple chemo-microscopical tests, and measuring cell sizes.
- **Chemical and physico-chemical aspects** — the chemical make-up of the crude drug, examined through qualitative and quantitative estimation of its constituents in the laboratory.
- **Surgical dressings and sutures** — identifying fibre types and checking their physical and chemical properties against pharmacopoeial standards.

Why pharmacognosy matters

- It confirms that crude drugs meet the required standards and are free from adulteration or substitution.
- It provides the preliminary information needed to identify, authenticate, and assess the quality of plant materials.
- Its diagnostic characters support accurate identification, help guide drug therapy, and feed into drug discovery and development.
- Above all, it ensures the *right* plant is used for its intended medicinal purpose.

General laboratory rules

- A clean, knee-length white laboratory coat must be worn at all times.
- No food or drink is allowed in the laboratory.

- All results must be recorded in your data book during the three-hour practical class — course-work assessment carries 40% of the total marks.
- Pre-study the practical schedule before every class.
- Any breakage of equipment is surcharged to the student responsible.
- Never wet the microscope stage with chemical reagents.
- Clean optical surfaces (lenses) only with special lens tissue paper.

Personal equipment you must bring
Trease & Evans *Pharmacognosy* textbook, a hand lens (×8 or ×10), a good-quality lead pencil (H or HB), razor blades, two dissecting needles, a medium spatula, a small water-colour brush, a soft eraser, and a transparent ruler (20 or 30 cm).

Chapter 2 • Use and Care of the Microscope

The microscope is the instrument that lets us examine structures far too small for the naked eye. First built by Zacharias Jansen around 1590, it magnifies fine detail so that cells, microorganisms, and other minute structures become visible — making it the single most important tool in practical pharmacognosy.

Parts of the microscope and what they do

Part	Function
Eyepiece / ocular lens	The lens at the top you look through; it further enlarges the image formed by the objective.
Nosepiece	Holds the objective lenses and rotates to switch between magnifications.
Arm	Supports the upper parts; grip it (with the other hand under the base) when carrying the microscope.
Stage	The flat platform that holds the slide, with a central opening to let light pass through.
Iris diaphragm	Controls how much light reaches the specimen, adjusting contrast and clarity.
Objective lenses	Capture light from the specimen, magnify it, and form a real, enlarged image.
Slide holder	Keeps the slide firmly in place on the stage while viewing.
Light source	Provides the illumination needed to view the specimen (a mirror or a built-in electric bulb).
Condenser	Focuses light from the source onto the specimen for clearer illumination.
Coarse focus knob	Moves the stage up or down quickly to locate the specimen before fine focusing.
Fine focus knob	Moves the stage slightly for precise, sharp focus — especially under high power.
Base	The heavy bottom that keeps the microscope steady and houses the light source.
On/off switch	Turns the light source on or off.
Inter-ocular adjustment	Adjusts the spacing between eyepieces to suit the viewer on a binocular microscope.

What the microscope is used for

- Observing microorganisms — bacteria, fungi, protozoa — and cells invisible to the naked eye.
- Examining blood, urine, and tissue samples to diagnose disease (malaria, tuberculosis, cancer cells).
- Studying DNA, cells, and microorganisms in detail across biotechnology, microbiology, and genetics.
- Allowing pathologists to inspect biopsy tissue and identify abnormal or cancerous cells.
- Analysing forensic evidence such as hair, fibres, and body fluids.
- Checking soil, water, and air samples for pollutants or microorganisms.

Using the microscope — step by step

- Settle into a comfortable sitting position with enough bench space for movement, specimens, and staining materials.
- Remove the plastic cover and make sure your allocated microscope is directly in front of you.
- Connect the electrical cable securely and switch on the light source.
- Place your slide on the stage, secure it with the object holder, and use the knurled knobs to move the stage.
- Set the nosepiece to the required objective — always start with low power.
- Move the stage until the object sits in the path of the light.
- Raise the stage to its uppermost position with the coarse focus, then lower it slowly until the image is sharp.
- Use the fine focus knob to sharpen the image after the coarse adjustment.
- When finished, lower the stage, switch off the power, and unplug the instrument.

Caring for the microscope

Mechanical parts: never attempt to clean, lubricate, or adjust the mechanical parts yourself. Report any fault or damage to the laboratory staff.

Optical parts — what NOT to do:

- Never let the lenses touch coverslips, slides, or the stage.
- Do not touch the lenses with your fingers — use lens tissue paper only.
- Never use alcohol on the lenses; it dissolves the cement that holds them together.
- Never dismantle an objective for internal cleaning.
- Make sure the undersides of slides are clean before mounting.
- Switch off the light when not in use, then cover the microscope and return it to its box.

Micrometry — measuring microscopic particles

Micrometry is the technique of measuring objects under the microscope. It matters in crude-drug work because it can distinguish a genuine drug from an adulterant that contains the same tissues but with cells of a different size. Measurement is done by comparing an object against reference lines in a graticule.

Tool	Description
Graticule (eyepiece micrometer)	A glass disc placed inside the eyepiece carrying an arbitrary scale, normally divided into 10 large and 100 small divisions.
Stage micrometer	A microscope slide bearing a 1 mm scale subdivided into 100 small divisions, each equal to 10 µm. It is used to calibrate the graticule.

Worked calibration examples

Low power: if 75 small graticule divisions line up with 100 small stage divisions (= 1 mm = 1000 µm), then 1 graticule division = 1000 ÷ 75 = **13.3 µm**.

High power: if 100 graticule divisions line up with 35 stage divisions (each stage division = 10 µm, so 350 µm), then 1 graticule division = 350 ÷ 100 = **3.5 µm**.

Chapter 3 • Practical Techniques in Pharmacognosy

Before a plant tissue can be identified under the microscope it usually has to be prepared — cleared of pigments, stained to reveal specific structures, and cut into thin sections. This chapter covers the reagents and methods used to do that.

A. Clearing agents for plant pigments

Clearing agent	Description and use
Chloral hydrate (50 g in 20 ml water)	Clarifies resistant structures by removing chlorophyll, starch, proteins, and mucilage — but **not** calcium oxalate or oils/fats. It also restores the original shape of cells in dried vegetable tissue.
Cresol	Penetrates the shell of prepared chalk to make its fine structure visible. Handle with care as it can irritate the skin; it does not need to be irrigated with glycerin.

B. Chemomicroscopical reagents (stains)

Cell-wall stains

Reagent	Action / colour
Iodine water (66 v/v) or 80% w/v H_2SO_4	Gives a blue-violet stain for cellulose cell walls.
Phloroglucinol (1% in 90% ethanol) + HCl	Stains lignified cell walls red. Let the phloroglucinol evaporate before adding 1–2 drops of concentrated HCl.
Tincture of Alkanna (alcoholic macerate of alkanet root)	Stains suberin (corky walls) and cutin (cuticularised walls) pink. Warm gently.
Corallin soda (0.25% in 25% aq. Na_2CO_3)	Stains the callose of phloem sieve tubes pink. Prepare fresh.
Ruthenium red BP (0.8% in 10% lead acetate)	Stains gums and mucilages red. Irrigate with 10% lead acetate.

Cell-content stains

Reagent	Action / colour
Iodine water	Stains starch blue-black and certain mucilages; proteins stain yellow-brown.
Picric acid (saturated)	Stains proteins in surgical-dressing fibres, or aleurone grains, yellow.
Tincture of Alkanna	Stains fixed oils, fats, and volatile oils red (warm gently).

C. Types of section

Two cuts give a clear picture of a specimen's internal structure:

- **Transverse section (T.S.)** — cut across the stem or root at right angles to the long axis.

- **Longitudinal section (L.S.)** — cut along the length of the organ.

D. Preparing specimens for microscopy

Staining procedure

- Rinse the sections in water and place them in a watch-glass. Add a few drops of phloroglucinol, then a drop of HCl.
- Remove the section from the stain and let the excess drain off for a few seconds.
- Transfer it into a drop of 50% glycerin on a clean slide and apply a coverslip.
- Lower the coverslip gently to avoid trapping air bubbles. Sections keep for at least a week in glycerin.

Leaf-surface preparation — scraping method

- Cut a suitable piece of leaf and lay it face down on a glazed tile or glass plate.
- Irrigate with a few drops of sodium hypochlorite and hold one end steady with a cork.
- Hold the blade at 90° to the leaf and scrape gently, keeping the leaf well irrigated.
- A thin, clear area forms which can be cut off; wash it in water.
- Remove any loosely adhering cells with a camel-hair paint brush.

E. Isolation techniques (macerating reagents)

These are needed to separate lignified tissues for microscopic examination. **All maceration must be done in a fume cupboard.**

- **Schultz method** (for grouped lignified elements): cover a small amount of drug with 2 ml of 50% nitric acid plus one crystal of potassium chlorate. Immerse the tube in a boiling water bath (controlled effervescence) and examine after 7–20 minutes under low power.
- **Isolation of simple lignified elements** (unlignified tissues): add fragments of drug to 2 ml of 5% w/v potassium hydroxide, heat in a boiling water bath for 10 minutes, and examine for disintegration.

Chapter 4 • Diagnostic Structures of Plant Material

Diagnostic characters are the specific, distinguishing features — morphological, genetic, or physiological — used to tell species apart. In pharmacognosy they are what let us confirm the identity of a drug and guarantee that the correct plant is being used.

1. Roots, rhizomes, and stems

Plant type	Structural feature
Dicotyledonous — primary state	Stems and roots show isolated vascular bundles, each with distinct areas of xylem, phloem, and cambium.
Dicotyledonous — secondary thickening	Parenchyma cells between the vascular bundles become meristematic again, producing secondary xylem and phloem.
Monocotyledonous plants	No secondary thickening; vascular bundles are scattered throughout the stem rather than arranged in a central ring.

2. Leaves

The leaf lamina shows one of two tissue-arrangement patterns:

- **Dorsiventral** (most leaves): upper epidermis → palisade parenchyma → spongy parenchyma → lower epidermis.

- **Isobilateral** (e.g. the T.S. of a Senna midrib): palisade tissue is present beneath *both* the upper and lower surfaces.

Leaf epidermis — key diagnostic features

Feature	Details
Cuticle	An outer layer of cutin (a fatty substance). It may be striated in some plants, which itself can be diagnostic.
Epidermal cells	A continuous surface layer. In stems and petioles they are elongated parallel to the long axis; in leaves and petals they have wavy anticlinal walls and are more or less isodiametric.
Stomata	Openings in the epidermis bounded by two specialised guard cells that control opening and closing.

Types of stomata

Type	Description	Example
Anomocytic	Stoma surrounded by a limited number of cells indistinguishable in size and shape from the other epidermal cells.	*Digitalis purpurea*
Anisocytic	Stoma surrounded by three or four subsidiary cells, one distinctly smaller than the others.	*Sedum*
Paracytic	Stoma flanked on each side by one or more subsidiary cells running parallel to the long axis of the pore.	*Vigna*
Diacytic	Stoma enclosed by two subsidiary cells whose common wall lies at right angles to the guard cells.	*Dianthus*

Trichomes (hair structures)

Trichomes are highly variable outgrowths of the epidermis. They include glandular (secretory) and non-glandular (covering) hairs, scales, papillae, and the absorbing hairs of roots. They can appear on any part of the plant and are strongly diagnostic for particular species.

- **Glandular trichomes** may have single or multicellular heads and stalks.
- **Non-glandular trichomes** may be single or multicellular, tufted, star-shaped, or branched.
- Some develop secondary walls with deposited lignin; others are marked by a warty cuticle.

3. Flowers

Commercially, a ‘flower’ may include whole inflorescences as well as individual flowers. The pericarp normally has three layers: the epicarp (outer epidermis), the endocarp (inner epidermis), and the mesocarp in between, which may be succulent or pithy.

4. Fruits

- **Simple fruits** — e.g. olive.
- **Aggregate fruits** — e.g. fennel (a cremocarp).
- **Compound fruits** — e.g. hops.

Umbelliferae fruits

Many fruits of pharmacognostic interest belong to the family **Umbelliferae** and are of one type — the **cremocarp**, a schizocarp (splitting fruit) that divides into two one-seeded portions called mericarps, each corresponding to one carpel. Fennel is the typical member; caraway, dill, coriander, anise, and hemlock are related.

5. Seeds

A seed develops from the fertilised ovule and contains the embryo. It is wrapped in a seed coat (the testa plus outer integument), within which lies a mass of storage tissue — the endosperm. The size of the embryo and the amount of endosperm vary widely from one seed type to another.

Chapter 5 • Macroscopy of Crude Vegetable Drugs

Macroscopic examination identifies a crude drug from its external features — shape, size, colour, texture, odour, taste, surface features, and arrangement. It is one of the simplest and most reliable methods of identification and a first line of defence against adulteration.

Examination format for each crude drug

For every specimen, record: common name (with vernacular name if known) · botanical name (genus, species, author) · family · morphological part · form (whole drug, powder, tincture) · major chemical constituents · a simple, *unshaded* sketch · and brief comments on what you observed.

Sample A — Bean (Phaseolus vulgaris)

Property	Details
Common name	Bean
Botanical name	*Phaseolus vulgaris*
Family	Fabaceae (Leguminosae) — the pea family
Morphological part	Seed coat, two cotyledons, embryo (hypocotyl + plumule), hilum, micropyle
Form	Whole drug — fresh, dried, or canned
Major constituents	15–30% protein, 60–70% carbohydrate, 0.7–2% lipid
Observations	0.5–2 cm long; flattened or kidney-shaped; white to dark brown; texture ranging hard to soft with moisture and variety
Commercially available	India, Brazil, Myanmar

Sample B — Turmeric (Curcuma longa)

Property	Details
Common name	Turmeric, yellow root (Yoruba: *Ateile pupo*)
Botanical name	*Curcuma longa* (genus *Curcuma*)
Family	Zingiberaceae
Morphological part	Rhizome — a thick, branching underground stem

Property	Details
Form	Fresh root (resembling ginger) and powdered spice
External characters	Cylindrical, irregular, often branched segments; orange-brown outer skin; rough, dry, slightly wrinkled surface with visible nodes and longitudinal ridges; mild characteristic aroma
Internal characters	Deep orange, slightly moist; yields a yellow-orange juice when pressed
Constituents & uses	Curcumin (the major active compound, gives the yellow colour), volatile oils (cinnamate compounds), proteins, starch, polyphenols. Curcumin is anti-inflammatory, antioxidant, and antimicrobial; used in capsules, tablets, and ointments

Sample C — Hibiscus flower (Hibiscus rosa-sinensis)

Property	Details
Common name	Rose mallow, red tea, Zobo (Nigeria: Zobo leaves)
Botanical name	*Hibiscus rosa-sinensis*
Family	Malvaceae
Placentation	Axile placentation
Floral formula	$C_5\ P_5\ A(\infty)\ G_{(5)}$
Description	Bisexual; 5 sepals; 5 petals at the base; androecium of numerous stamens; gynoecium of 5 fused carpels (a 3-carpous ovary)
Constituents	Flavonoids, anthocyanins, amino acids, alkaloids, saponins

Sample D — False Ashoka leaves (Polyalthia longifolia)

Property	Details
Common name	False Ashoka, masquerade tree, mast tree, Indian fir tree
Botanical name	*Polyalthia longifolia*
Family	Annonaceae
Description	Glossy green, smooth, leathery leaves; smooth margins; pinnate venation; acuminate apex; slightly cordate base
Arrangement / shape	Alternate phyllotaxy; lanceolate (narrow and elongated)
Constituents	Alkaloids, tannins, flavonoids, saponins — anti-inflammatory, antimicrobial, antioxidant

Papaw versus Okra

Property	Papaw (Sample 1)	Okra (Sample 2)
Botanical name	*Carica papaya*	*Abelmoschus esculentus*
Common name	Pawpaw (Yoruba: *Ibepe*)	Okra / lady's finger (Yoruba: *Ila*)
Family	Caricaceae	Malvaceae

Property	Papaw (Sample 1)	Okra (Sample 2)
Description	Large, oval to oblong; smooth green outer surface. Inside: soft fleshy mesocarp with numerous small black seeds in a gelatinous covering	Elongated, green, ridged, tapering. Inside: a central core with rows of white round seeds embedded in mucilage
Placentation	Axile / parietal	Axile
Seed observation	Small, black seeds with a transparent mucilaginous layer; press to release fluid (oxalate present)	Round, soft seeds; press to release a sticky mucilaginous substance (oxalate present)

Chapter 6 • Study of Starches (Schedule II)

Starches are polysaccharides that yield only glucose on complete hydrolysis, so they sit under the general heading of carbohydrates. Hot water splits starch into an insoluble fraction (amylopectin) and a soluble fraction (amylose); the amylose is then hydrolysed exclusively to maltose by the enzyme β-maltase.

Sources of pharmaceutical starch

Plant source	Botanical name
Plantain / banana	*Musa paradisiaca*
Wheat	*Triticum aestivum*
Breadfruit	*Artocarpus altilis*
Cassava	*Manihot utilissima*
Maize (corn)	*Zea mays*
Potato	*Solanum tuberosum*

Pharmaceutical uses of starch

- As a diluent in tablet manufacture.
- As an indicator in iodometric analyses.
- As a tablet excipient — disintegrant, filler, and binder.
- As an antidote in iodine poisoning.
- In dusting powders, and for starching and sizing fabrics.

Methods

- **Microscopic test:** place a little starch on a slide, moisten with alcohol, add a drop of dilute glycerol, and stir gently. Apply a coverslip and examine. Add dilute iodine solution and watch for the colour change that confirms starch.
- **Macroscopic (solubility) test:** transfer 1 g of starch into 10 ml of cold water, stir, and note the result. Then heat gently to boiling, allow to cool, and record further observations.

Results — the three starches compared

Feature	Maize (*Zea mays*)	Potato (*S. tuberosum*)	Wheat (*T. aestivum*)
Family	Poaceae	Solanaceae	Poaceae

Feature	Maize (*Zea mays*)	Potato (*S. tuberosum*)	Wheat (*T. aestivum*)
Macroscopic	White, odourless, smooth, fine, tasteless	White, odourless, tasteless; fine powder	Light brown, tasteless, smooth, slightly gritty
Solubility (cold)	Insoluble	Insoluble	Insoluble
Solubility (hot)	Forms a thick gelatinous paste	Highly gelatinised, thick paste (high amylopectin)	Forms a gelatinous paste
Granule shape	Small, angular or polygonal	Large, oval	Round (mixture of large and small)
Hilum	Central hilum (dot)	Distinct hilum	Eccentric hilum
Iodine test	Blue-black	Blue-black	Blue-black

Key takeaways

All three starches are **insoluble in cold water** but **gelatinise in hot water**, and all give a **blue-black** iodine reaction. The differences that matter are in granule shape and hilum — and those differences are exactly what let you detect adulteration in a pharmaceutical preparation.

Chapter 7 • Unorganised Crude Drugs: Gums (Schedule III)

Unorganised crude drugs are natural medicinal substances that lack a definite cellular structure; they are usually plant exudates or secretions. Their chief constituents are gums, mucilages, resins, and oils. Gums themselves are heterogeneous polysaccharides the plant produces to cover accidental wounds and reduce water loss — they absorb water and swell, which makes them valuable formulation aids.

Classification of unorganised crude drugs

Type	Example	Type	Example
Dried latex	Opium	**Oleo-resins**	Copaiba
Dried juice	Aloes	**Waxes**	Beeswax
Extracts	Catechu	**Oils and fats**	Castor oil
Gums	Acacia	**Volatile oil**	Clove oil, *Eugenia uniflora*
Resins	Colophony	**Balsams**	Balsam of Tolu
Gum-resins	Myrrh	**Rubber**	*Hevea brasiliensis*

Sample A — Tragacanth gum (Astragalus gummifer)

- **Source:** the natural dried gummy exudate from the stems and branches of *Astragalus gummifer*.
- **Appearance:** white to yellowish flakes or ribbon-like pieces. In water it swells to a thick gel but does NOT fully dissolve.
- **Pharmaceutical uses:** suspending, emulsifying, and thickening agent; also a tablet binder.

Sample B — Acacia gum / gum arabic (Acacia senegal)

- **Source:** the dried gummy exudate from the stems and branches of *Acacia senegal*.

- **Appearance:** pale yellow to brownish glassy tears. Dissolves readily in water to give a viscous solution.
- **Pharmaceutical uses:** emulsifying agent, binder, and stabiliser; used in lozenges and syrups.

General tests

Test	Tragacanth (*Astragalus*)	Acacia (*A. senegal*)
Physical appearance	Pale yellow; odourless, tasteless; rough and lumpy	White to brownish; smooth fine powder; bland
Solubility (cold water)	Does not fully dissolve; swells into a gel	Dissolves readily to a clear / opalescent solution
Solubility (alcohol)	Insoluble	Insoluble
Flocculation (drug + water)	Sticky mucilage forms	Sticky mucilage forms
Drug + water + alcohol	Solution becomes more sticky	Solution becomes more sticky

Identification tests

Reagent / method	Tragacanth (inference)	Acacia (inference)
0.2% lead acetate	White milky precipitate (gum present)	White milky precipitate (gum present)
N/50 iodine solution	Golden brown / turns red (starch absent)	Dark purple (starch impurity present)
10% $FeCl_3$ solution	Deep yellow precipitate (gum / mucilage)	No visible reaction (tannins absent)

Conclusion — Acacia vs Tragacanth (a favourite exam question)
• Acacia is **completely water-soluble**; Tragacanth only **swells**.
• Iodine test: Acacia → **dark purple**; Tragacanth → **golden brown**.
• $FeCl_3$: Acacia → **no reaction** (tannins absent); Tragacanth → **deep yellow precipitate**.
• Both are insoluble in alcohol and both give a white precipitate with lead acetate.

Chapter 8 • Examination of Fibres and Surgical Dressings (Schedule I)

Identifying fibres matters because many are used in surgical dressings, ligatures, and sutures — while others, such as human or rodent hair, can be dangerous contaminants in sterile products. Surgical dressings are evaluated as textile products against pharmacopoeial standards.

Surgical dressings — quality standards assessed

- Level of adhesiveness (mainly for plasters).
- Area per unit weight, and weight of fabric.
- Colour fastness.
- Elasticity and extensibility.
- Foreign matter and sterility.
- Threads per stated length.
- Absorbency, water-soluble extractives, and water-vapour permeability.

Chemical evaluation results (absorbent cotton wool & gauze)

Test / reagent	Observation	Inference
4.5% NaOH (5 min heat)	Wool: translucent / gelatinous. Gauze: swollen, softer texture	Partial, incomplete dissolution — alkali breaks hydrogen bonds and raises water absorption
Dilute H_2SO_4	Wool and gauze: no visible change	No dissolution — high chemical stability under mild acid
50% w/w H_2SO_4	Wool: gradual swelling. Gauze: gradual thinning	Partial dissolution and progressive degradation
70% H_2SO_4	Wool: thick, slightly dark gelatinous solution. Gauze: clear, slightly viscous solution	Complete dissolution — confirms both are purely cellulosic

Absorbency and elasticity results

Test	Result / standard
Absorbency (1.5 g cotton wool)	Time to sink = 8 seconds. Meets the BPC standard (≤ 10 seconds) for efficient absorption of wound exudate
Elasticity — unstretched length	10 cm
Elasticity — fully stretched	28 cm (meets BPC: must be ≥ twice the unstretched length)
Elasticity — regained length	12 cm (meets BPC: must be ≤ two-thirds of the fully stretched length)

Why the acid results matter

The graded response to sulphuric acid tells the whole story: dilute acid does nothing, 50% acid partly degrades the fibre, and 70% acid dissolves it completely. Complete dissolution in strong acid is the confirmation that a surgical dressing is **pure cellulose**.

Chapter 9 • Diagnostic Characters of Medicinal Plants

This practical compares two common medicinal plants — *Tithonia diversifolia* (Mexican sunflower) and *Talinum triangulare* (water leaf) — using their macroscopic and microscopic features. The two make an ideal teaching pair because almost every character differs between them.

Sample 1 — Tithonia diversifolia (Mexican sunflower)

- **Macroscopic:** large, rough, ovate leaf with a serrated margin; rough surface bearing noticeable hairs; reticulate venation; a characteristic smell when crushed.
- **Microscopic (leaf):** numerous uniseriate and multicellular trichomes; anomocytic stomata; a thick cuticle; prominent, well-developed vascular bundles.

Stem-bark T.S. feature	Observation
Calcium oxalate crystals	Prismatic crystals, numerous in the phloem
Lignified fibres	Long, thick-walled, with a narrow lumen; stain deep red; arranged in bundles within the phloem
Scleroids (stone cells)	Small lumen; isodiametric, thick-walled cells with visible pits
Vascular bundle	A distinct ring of vascular bundles

Sample 2 — Talinum triangulare (water leaf)

- **Macroscopic:** smooth, succulent, broad leaf with an entire margin; soft, watery texture; venation not prominent (because of the succulence); no characteristic hairs.

Leaf feature	Observation
Epidermis	Thin-walled, polygonal cells; stomata abundant on the lower surface
Stomata	Paracytic
Calcium oxalate crystals	Raphide (needle-like) crystals, obvious in the palisade
Trichomes	Generally absent
Other	Mucilage-containing cells present — characteristic of succulents

Side-by-side comparison

Feature	*Tithonia diversifolia*	*Talinum triangulare*
Texture	Rough, large, ovate; serrated margins	Smooth, soft, succulent, broad; entire margins
Trichomes	Numerous uniseriate, multicellular (abundant on both surfaces)	No noticeable trichomes
Stomata	Anomocytic; mostly on the lower epidermis	Paracytic; visible on both surfaces
Cuticle	Thick, on elongated epidermal cells	Thin compared to Tithonia
Calcium oxalate	Prismatic crystals (occasionally cluster/druse)	Raphide crystals; obvious in the palisade
Mucilage cells	Not prominent	Present — characteristic of succulents
Vascular bundles	Prominent and well-developed	Not very prominent (succulence)
Lignified fibres	Long, thick-walled, deep red	Thin to moderately thick; less lignified

The single most important distinction

***Tithonia* has abundant trichomes; *Talinum* has none.** If both plants were dried and powdered, these microscopic characters would be the only reliable way to prevent substitution or adulteration.

Chapter 10 • Vernonia amygdalina (Bitter Leaf)

Vernonia amygdalina — bitter leaf — is a member of the family Asteraceae. Its stem bark shows a distinctive set of microscopic characters that together confirm the powdered sample really comes from the leaf of this species.

Feature	Observation
Calcium oxalate crystals	Both prismatic and druse (cluster) crystals present
Vessels and fibres	Spiral and annular xylem vessels present
Epidermal cells	Polygonal to irregular; a clear distinction between upper and lower epidermis; moderately thick walls
Stomata	Paracytic (rubiaceous type); mostly hypostomatic — on the lower epidermis

Feature	Observation
Trichomes	Multicellular, uniseriate, non-glandular, straight; more abundant on the lower epidermis

Taken together — the two crystal types, the paracytic stomata, and the uniseriate trichomes on the lower surface — these features conclusively identify the sample as the leaf of *Vernonia amygdalina*.

Chapter 11 • Mineral Matters (Schedule IV)

Most mineral matters used in pharmacognosy occur as natural deposits in soil from different regions. Some are plain inorganic compounds; others carry organic components or a fossilised origin. This schedule examines four crude drugs of mineral origin — kaolin, bentonite, chalk, and talc — looking at their organoleptic properties, microscopic structure, physical behaviour, and pharmaceutical use.

How to tell the four apart at a glance

Kaolin settles at a moderate rate and doesn't swell. **Bentonite** hardly settles — it swells 12–15× into a gel. **Chalk** settles fast and shows fossil shells. **Talc** is intensely slippery and floats before slowly settling. The *sedimentation behaviour* is the quickest discriminator.

Specimen 1 — Kaolin (hydrated aluminium silicate)

Aspect	Details
Organoleptic	White to grayish-white fine powder; smooth, highly unctuous (soapy) and plastic when moistened; earthy/clay taste. On the palm it leaves a smooth, slightly drying opaque white film that adheres well without grittiness
Microscopy	Mounted in cresol or chloral hydrate. Appears as irregular, angular, flat, polymorphic particles — small plate-like shards rather than a definite crystalline form
Sedimentation (1 g / 200 ml)	Light kaolin suspends well at first as a cloudy dispersion, then settles at a moderate rate; it does not swell significantly. Heavy kaolin settles much faster
Applications	Adsorbent for GI toxins (antidiarrhoeal); external dusting powder, poultices, and face masks; diluent and excipient in tablets
Products	Kaopectate (historical antidiarrhoeal); Kaolin Poultice BP (reduces inflammation)

Specimen 2 — Bentonite (colloidal hydrated aluminium silicate)

Aspect	Details
Organoleptic	Pale buff, cream, or grayish powder; smooth to the touch; earthy taste. Rubbed dry it feels smooth — add a drop of water and it becomes highly sticky and slippery as it swells
Microscopy	Mounted in cresol; shows irregular, fine granular particles. Unlike chalk it lacks fossilised structures
Sedimentation (1 g / 200 ml)	Does not readily sediment. Instead it absorbs water and swells to about 12–15× its dry volume, forming a highly viscous, gel-like colloidal dispersion
Applications	Suspending agent for insoluble powders in lotions and mixtures; stabiliser in oil-in-water emulsions; ointment and plaster base
Products	Calamine lotion (as the suspending agent); topical gels and barrier creams

Specimen 3 — Chalk (prepared calcium carbonate)

Aspect	Details
Organoleptic	Pure white to slightly off-white; slightly rough or micro-gritty (prepared chalk); precipitated chalk is finer; tasteless. Leaves a dry, chalky residue that adheres less strongly than kaolin or talc
Microscopy	Mounted in cresol (which penetrates the shell to reveal fine internal structure). Prepared chalk shows the fossilised shells of unicellular sea organisms (Foraminifera) — small rounded or geometric fossil shapes. Precipitated chalk lacks these and shows microcrystalline aggregates
Sedimentation (1 g / 200 ml)	Practically insoluble; settles relatively quickly compared with bentonite or light kaolin, leaving the supernatant clear
Applications	Antacid to neutralise stomach acid; calcium supplement; mild abrasive in dental preparations
Products	Tums / Rennie (antacid chewables); toothpastes (as a polishing abrasive)

Specimen 4 — Talc (purified magnesium silicate)

Aspect	Details
Organoleptic	Pure white to grayish-white, very fine powder; extremely unctuous, slippery, and greasy to the touch (despite containing no fat); tasteless. On the palm it gives extreme 'slip' and leaves a thin, smooth, cooling coating
Microscopy	Mounted in chloral hydrate; appears as irregular, distinctly foliated (plate-like) or lamellar crystals that often overlap
Sedimentation (1 g / 200 ml)	Highly hydrophobic — much of it floats at first; with agitation it slowly disperses and eventually settles without swelling
Applications	Glidant and lubricant in tablet manufacture (improves flow, prevents sticking); dusting powder to prevent chafing; clarifying agent for liquids
Products	Johnson's Baby Powder (historically); a lubricant/glidant in almost all solid oral dosage forms (e.g. paracetamol, aspirin tablets)

Macroscopic examination of proprietary drug products (Schedule V)

This schedule extends macroscopy to commercial pharmaceutical products alongside natural crude drugs. For each of five proprietary products sold in registered pharmacies, record: the registered trade-mark name; the active chemical constituents (e.g. paracetamol, caffeine, acetylsalicylic acid) and other excipients (e.g. kaolin); and the natural source of both the actives and the excipients, using standard textbooks. For the fruit specimens re-examined here — papaw and okra — the constituents and relevance are:

Constituent property	Papaw (*Carica papaya*)	Okra (*Abelmoschus esculentus*)
Constituents	Papain, carotenoids, alkaloids, vitamin C, flavonoids, saponins	Mucilage, carbohydrates, proteins, vitamin C, flavonoids, polyphenols
Pharmaceutical relevance	Papain aids digestion and wound healing; carotenoids and vitamin C act as antioxidants	Mucilage acts as a demulcent and thickening agent for cough syrups and tablets; flavonoids and polyphenols are antioxidants

Chapter 12 • Quick Revision — Key Facts for the MCU

Essential reagents and their reactions

Reagent	What it tests / does	Result / colour
Iodine (N/50 or dilute)	Tests for starch; identifies gum type	Starch: blue-black. Acacia: dark purple. Tragacanth: golden brown
0.2% lead acetate	Identification test for gums	White milky precipitate = gum present
10% $FeCl_3$	Tests for tannins and gum mucilage	Acacia: no reaction (tannins absent). Tragacanth: deep yellow precipitate
4.5% NaOH	Tests cellulose fibres	Gelatinous swelling — partial dissolution
50% H_2SO_4	Moderate acid on cellulose	Gradual swelling — partial dissolution
70% H_2SO_4	Strong acid hydrolysis of cellulose	Complete dissolution — confirms pure cellulose
Phloroglucinol + HCl	Tests for lignified cell walls	Red staining of lignified cells
Ruthenium red	Tests for gums and mucilages	Red staining
Iodine + 80% H_2SO_4	Tests for cellulose cell walls	Blue-violet stain

Key botanical identifications

Plant	Family	Key diagnostic feature
Tithonia diversifolia	Asteraceae	Numerous multicellular trichomes; anomocytic stomata; prismatic calcium oxalate crystals
Talinum triangulare	Portulacaceae	No trichomes; paracytic stomata; raphide crystals; mucilage cells
Vernonia amygdalina	Asteraceae	Prismatic + druse calcium oxalate; paracytic stomata; multicellular trichomes on the lower surface
Hibiscus rosa-sinensis	Malvaceae	Axile placentation; floral formula $C_5P_5A(\infty)G_{(5)}$
Polyalthia longifolia	Annonaceae	Lanceolate leaves; alternate phyllotaxy; leathery texture
Carica papaya	Caricaceae	Axile/parietal placentation; numerous seeds with a gelatinous covering
Abelmoschus esculentus	Malvaceae	Mucilaginous seeds; axile placentation
Phaseolus vulgaris	Fabaceae	15–30% protein; hilum and micropyle present
Curcuma longa	Zingiberaceae	Orange rhizome; curcumin as the major constituent
Acacia senegal	Fabaceae	Dissolves in water; insoluble in alcohol; no reaction with $FeCl_3$
Astragalus gummifer	Fabaceae	Swells but does not fully dissolve; deep yellow precipitate with $FeCl_3$

Starch granules at a glance

Starch	Shape	Hilum	Size
Maize (*Zea mays*)	Small, angular / polygonal	Central dot	Small
Potato (*S. tuberosum*)	Large, oval (egg-shaped)	Distinct, eccentric	Largest of the three

Starch	Shape	Hilum	Size
Wheat (*T. aestivum*)	Round; large and small mixed	Eccentric	Mixed

BPC standards — surgical dressings

Parameter	BPC specification
Absorbency (cotton-wool sinking time)	Not more than 10 seconds
Elasticity — fully stretched length	Not less than twice the unstretched length
Elasticity — regained length	Not more than two-thirds of the fully stretched length

Top 10 MCU tips

1. Know the difference between **organised** (definite cellular structure) and **unorganised** (no cellular structure) crude drugs.
2. The iodine test always gives **blue-black** for starch across all types (maize, potato, wheat).
3. **Acacia dissolves readily** in water; **Tragacanth swells** but does not fully dissolve — the key differentiator.
4. **70% H_2SO_4** causes complete dissolution of cellulose fibres, confirming pure cellulose.
5. **Tithonia has abundant trichomes; Talinum has none** — the most important visual distinction.
6. Stomata types: anomocytic (Tithonia, Digitalis), paracytic (Talinum, Vernonia), plus anisocytic and diacytic.
7. Calcium oxalate: **raphides** (needle-like) in Talinum; **prismatic** in Tithonia and Vernonia; **druse/cluster** in Vernonia.
8. Starch granules: potato → large oval; maize → small angular; wheat → round, mixed sizes.
9. Curcumin (from *Curcuma longa*) gives turmeric its yellow colour and is anti-inflammatory, antioxidant, and antimicrobial.
10. BPC: cotton-wool sinking time ≤ 10 seconds; stretched bandage ≥ twice the unstretched length.

Final word

Review each chapter, focus on the results tables, and make sure you can recall the botanical names, families, and key diagnostic features for every specimen. **Good luck with your MCU test!**

Originality & IP Disclaimer

This study guide consolidates the second-semester Pharmacognosy practical syllabus into a single original revision resource. It is shared freely for students — please do not sell it.

everythingabuad.com